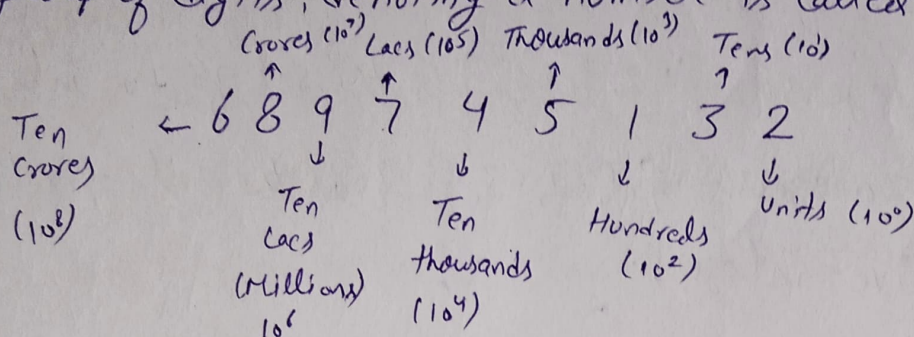


Numbers

(1)

Numerals:- In Hindu Arabic system, we use ten symbols 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 called digits to represent any number.

→ A group of digits, denoting a number is called a numeral.



Place Value or Local Value of a Digit in a Numeral.

In the above Numeral:

Place Value of 2 is $(2 \times 1) = 2$

Place Value of 3 is $(3 \times 10) = 30$

Place Value of 1 is $(1 \times 100) = 100$

Place Value of 6 is $(6 \times 10^8) = 600000000$

Face Value:- The face value of a digit in a numeral is the value of the digit itself at whatever place it may be.

In the above numeral, the face value of 2 is 2, the face value of 3 is 3 and so on.

Types of Numbers:-

1. Natural Numbers:- Counting numbers 1, 2, 3, 4, 5, ... are called Natural numbers.

2. Whole Numbers:- All counting numbers together with zero are Whole Numbers.

Note (i) 0 is the only whole number which is not a natural Number.

(ii) Every Natural number is a whole number.

3. Integers:- All natural numbers, 0 and negatives of counting numbers i.e. $\dots -3, -2, -1, 0, 1, 2, 3, \dots$ together form the set of integers.

(i) Positive Integers:- $\{1, 2, 3, \dots\}$ is the set of +ve integers

(ii) Negative Integers:- $\{-1, -2, -3, \dots\}$ is the set of all negative integers.

(iii) Non Positive and Non-Negative Integers:- 0 is neither positive nor negative. So $\{0, 1, 2, 3, \dots\}$ represents the set of non negative integers, while $\{0, -1, -2, \dots\}$ represents the set of non-positive integers.

4. Even Numbers:- A number divisible by 2 is called an even number. Ex:- 2, 4, 6, 8, 10, $\dots$ etc

5. Odd Numbers:- A number not divisible by 2 is called an odd number. Ex:- 1, 3, 5, 7, 9, 11, $\dots$ etc.

6. Prime Numbers:- A number greater than 1 is called a prime number, if it has exactly two factors, namely 1 and the number itself.

Note:- Prime Numbers upto 100 are: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97.

Prime Number Greater than 100:-

(2)

Let p be a given number greater than 100. To find out whether it is prime or not, we use the following method:

Find a whole number nearly greater than the square root of p . Let $k > \sqrt{p}$. Test whether p is divisible by any prime number less than k . If yes, then p is not prime. Otherwise p is prime.

Ex:- We have to find whether 191 is a prime number or not. Now, $14 > \sqrt{191}$.

Prime numbers less than 14 are 2, 3, 5, 7, 11, 13.

191 is not divisible by any of them. So, 191 is a prime number.

7. Composite Numbers:- Numbers greater than 1 which are not prime, are known as composite numbers.

Ex:- 4, 6, 8, 9, 12, 10, ...

Note:- (i) 1 is neither prime nor composite.

(ii) 2 is the only even prime number which is prime.

(iii) There are 25 prime numbers between 1 and 100.

8. Co-Primes:- Two numbers a and b are said to be co-primes, if their H.C.F. is 1.

Ex:- (2, 3), (4, 5), (7, 9), (8, 11), etc. are co-primes.

9. Twin Primes:- A twin prime is a prime number that is either 2 less or 2 more than another prime number. Ex:- (3, 5), (5, 7), (11, 13), (17, 19).

* Tests of Divisibility:-

1. Divisibility by 2: A number is divisible by 2, if its unit's digit is any of 0, 2, 4, 6, 8.

Ex:- 84932 is divisible by 2, while 65935 is not.

2. Divisibility by 3:- A number is divisible by 3, if the sum of its digits is divisible by 3.

3. Divisibility by 4:- A number is divisible by 4, if the number formed by the last two digits is divisible by 4.

Ex:- 892648 is divisible by 4, since the number formed by the last two digits is 48, which is divisible by 4.

But, 749282 is not divisible by 4, since the number formed by the last two digits is 82, which is not divisible by 4.

4. Divisibility by 5:- A number is divisible by 5, if its unit's digit is either 0 or 5. Thus, 20820 and 50345 are ~~div~~ divisible by 5, while 30934 and 40946 are not.

5. Divisibility by 6:- A number is divisible by 6, if it is divisible by both 2 and 3.

Ex:- The number 35256 is clearly divisible by 2.

Sum of its digits = $(3 + 5 + 2 + 5 + 6) = 21$ which is divisible by 3.

35256 is divisible by 2 as well as 3. Hence, 35256 is divisible by 6. (3)

Divisibility by 8:- A number is divisible by 8, if the number formed by the last three digits of the given number is divisible by 8.

Ex:- 953360 is divisible by 8, since the number formed by last three digits is 360, which is divisible by 8.

But 529418 is not divisible by 8, since the number formed by last three digits is 418, which is not divisible by 8.

Divisibility by 9:- A number is divisible by 9, if the sum of its digits are divisible by 9.

Ex:- 60732 is divisible by 9, since sum of digits = $(6+0+7+3+2) = 18$, which is divisible by 9.

But 68956 is not divisible by 9,

$$\text{as } 6+8+9+5+6 = 34.$$

Divisibility by 10:- A number is divisible by 10, if it ends with 0.

Ex:- 96410, 10480 are divisible by 10.

Divisibility by 11:- A number is divisible by 11, if the

difference of the sum of its digits at odd places and the sum of its digits at even places, is either 0 or a number divisible by 11.

The number 4832718 is divisible by 11, since:

$$(\text{sum of digits at odd places}) - (\text{sum of digits at even places}) \\ = (8 + 7 + 3 + 4) - (1 + 2 + 8) = 11, \text{ which} \\ \text{which is divisible by 11.}$$

Divisibility by 12:- A number is divisible by 12, if it is divisible by both 4 and 3.

Ex:- Consider the number 34632.

- i) The number formed by last two digits is 32, which is divisible by 4.
- ii) Sum of digits $= (3 + 4 + 6 + 3 + 2) = 18$, which is divisible by 3.

Divisibility by 14:- A number is divisible by 14, if it is divisible by 2 as well as 7.

Divisibility by 15:- A number is divisible by 15, if it is divisible by both 3 and 5.

Divisibility by 16:- A number is divisible by 16, if the number formed by the last 4 digits is ~~divib~~ divisible by 16.

Ex:- 7957536 is divisible by 16, since the number formed by the last four digits is 7536, which is divisible by 16.

Divisibility by 24:- A given number is divisible by 24, if it is divisible by both 3 and 8.

(4)

Divisibility by 40 :- A given number is divisible by 40, if it is divisible by both 5 and 8.

Divisibility by 80 :- A given number is divisible by 80, if it is divisible by both 5 and 16.

Note :- If a number is divisible by p as well as q , where p and q are co-primes, then the given number is divisible by pq .

If p and q are not co-primes, then the given number need not be divisible by pq even when it is divisible by both p and q .

Ex :- 36 is divisible by 4 and 6, but it is not divisible by $(4 \times 6) = 24$, since 4 and 6 are not co-primes.

* Multiplication by Short Cut Methods :-

(i) @ Multiplication by Distributive Law :-

$$a \times (b + c) = a \times b + a \times c$$

$$(ii) \quad a \times (b - c) = a \times b - a \times c.$$

Ex :-

$$\begin{aligned} 567958 \times 99999 &= 567958 \times (100000 - 1) \\ &= (56795800000 - 567958) \\ &= 56795232042. \end{aligned}$$

$$\begin{aligned} (ii) \quad 978 \times 184 + 978 \times 816 &= 978(184 + 816) \\ &= 978 \times 1000 = 978000. \end{aligned}$$

Multiplication of a Number by 5^n . Put n zeros to the right of the multiplicand and divide the number formed by 2^n .

Ex:- $975436 \times 625 = 975436 \times 5^4 = \frac{9754360000}{16}$
 $= 609647500.$

* Basic Formulae:-

1. $(a+b)^2 = a^2 + b^2 + 2ab$
2. $(a-b)^2 = a^2 + b^2 - 2ab$
3. $(a+b)^2 - (a-b)^2 = 4ab$
4. $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$
5. $(a^2 - b^2) = (a+b)(a-b)$
6. $(a+b+c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$
7. $(a^3 + b^3) = (a+b)(a^2 - ab + b^2)$
8. $(a^3 - b^3) = (a-b)(a^2 + ab + b^2)$
9. $(a^3 + b^3 + c^3 - 3abc) = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$
10. If $a+b+c = 0$, then $a^3 + b^3 + c^3 = 3abc$.

* Division Algorithm or Euclidean Algorithm:-

If we divide a given number by another number, then:

$$\text{Divident} \quad \text{Dividend} = (\text{Divisor} \times \text{Quotient}) + \text{Remainder.}$$

- (i) $(x^n - a^n)$ is divisible by $(x-a)$ for all values of n .
- (ii) $(x^n - a^n)$ is divisible by $(x+a)$ for all ^{even} values of n .
- (iii) $(x^n + a^n)$ is divisible by $(x+a)$ for all odd values of n .

Progression:- A succession of numbers formed and arranged in a definite order according to certain definite rule, is called a progression.

1. Arithmetic Progression (A.P.) If each term of a progression differs from its preceding term by a constant, then such a progression is called an arithmetical progression. This constant difference is called the common difference of the A.P. An A.P. with first term 'a' and common difference d is given by $a, (a+d), (a+2d), (a+3d), \dots$

The n^{th} term of this A.P. is given by $T_n = a + (n-1)d$.

The sum of n terms of this A.P.

$$S_n = \frac{n}{2} [2a + (n-1)d] = \frac{n}{2} [\text{first term} + \text{last term}]$$

Some Important Results:-

(i) $(1 + 2 + 3 + \dots + n) = \frac{n(n+1)}{2}$

(ii) $(1^2 + 2^2 + 3^2 + \dots + n^2) = \frac{n(n+1)(2n+1)}{6}$

(iii) $(1^3 + 2^3 + 3^3 + \dots + n^3) = \frac{n^2(n+1)^2}{4}$

2. Geometrical Progression (G.P.):- A progression of numbers in which every term bears a constant ratio with its preceding term, is called a geometrical progression.

The constant ratio is called the common ratio of the G.P.

A G.P. with first term a and common ratio r is $a, ar, ar^2, ar^3, \dots$

In this G.P. $T_n = a \cdot r^{n-1}$.

Sum of the n terms, $S_n = \frac{a(1-r^n)}{(1-r)}$.

Q:1 What value will replace the question mark in each of the following equations?

(i) $? - 1936248 = 1635773$

(Ans - 3572021)

(ii) $8597 - ? = 7429 - 4358$

(Ans - 5526)

Q:2 What could be the maximum value of Q in the following equation?

$5P9 + 3R7 + 2Q8 = 1114$

We may analyse the given eqn as

$2 + P + R + Q = 11$

①	②
5	P 9
3	R 7
2	Q Q 8
1114	

So the maximum value of Q can be

$(11 - 2)$ i.e. 9 (when $P = 0, R = 0$).

Q:3 Simplify : (i) 5793405×9999 (ii) 839478×625

Solⁿ (i) $5793405 \times 9999 = 5793405 (10000 - 1)$
 $= 57934050000 - 5793405$
 $= 57928256595$

(6)

$$839478 \times 625 = 839478 \times 54 = \frac{839478000}{16} = 524673750$$

Ex-5 Evaluate: (i) $986 \times 137 + 986 \times 863$

$$= 986(137 + 863) = 986 \times 1000 = 986000$$

(ii) $983 \times 207 - 983 \times 107 = 983(207 - 107) = 983 \times 100 =$

Ex-6 Simplify (i) 1605×1605 (ii) (1398×1398)

Solⁿ $1605 \times 1605 = (1605)^2 = (1600 + 5)^2$

$$= (1600)^2 + (5)^2 + 2 \times 1600 \times 5$$

$$= 2560000 + 25 + 16000 = 2576025.$$

(ii) $(1398 \times 1398) = (1398)^2 = (1400 - 2)^2$

$$= (1400)^2 + (2)^2 - 2 \times 1400 \times 2$$

$$= 1960000 + 4 - 5600 = 1954404.$$

Ex-7 Evaluate:- $313 \times 313 + 287 \times 287$

Solⁿ $(a^2 + b^2) = \frac{1}{2} [(a+b)^2 + (a-b)^2]$

$$\therefore (313)^2 + (287)^2 = \frac{1}{2} [(313 + 287)^2 + (313 - 287)^2]$$

$$= \frac{1}{2} [(600)^2 + (26)^2]$$

$$= \frac{1}{2} [360000 + 676]$$

$$= 180338.$$

Ex:-8 Which of the following are prime numbers?

- (i) 241 (ii) 337 (iii) 391 (iv) 571

Solⁿ Clearly, ~~16~~ $16 > \sqrt{241}$

Prime numbers less than 16 are 2, 3, 5, 7, 11, 13

241 is not divisible by any one of them

$\therefore$ 241 is a prime number.

(ii) $19 > \sqrt{337}$

Prime numbers less than 19 are 2, 3, 5, 7, 11, 13, 17.

337 is not divisible by any one of them.

337 is a prime number.

(iii) Clearly $20 > \sqrt{391}$.

Prime numbers less than 20 are 2, 3, 5, 7, 11, 13, 17, 19.

We find that 391 is divisible by 17.

391 is not prime.

(iv) Clearly, $24 > \sqrt{571}$.

Prime numbers less than 24 are 2, 3, 5, 7, 11, 13, 17, 19, 23.

571 is not divisible by any one of them.

$\therefore$ 571 is a prime number.

Ex:-9 Find the unit's digit in the product $(2467)^{153} \times (341)^{72}$

Solⁿ Clearly, Unit's digit in the given product =

Unit's digit in $7^{153} \times 1^{72}$

$\therefore 7^4$ gives unit digit 1.

$\therefore 7^{152}$ gives unit digit 1.

$\therefore 7^{153}$ gives the unit digit $(1 \times 7) = 7$.

Also 1^{72} gives unit digit 1.

Hence, unit's digit in the product $= (7 \times 1) = 7$.

Ex:-10 Find the unit's digit in $(264)^{102} + (264)^{103}$

Solⁿ Required unit digit = Unit digit in $(4)^{102} + (4)^{103}$

Now, 4^2 gives unit digit 6.

$(4)^{102}$ gives unit digit 6.

$(4)^{103}$ gives unit digit (6×4) i.e. 4

Hence, unit digit in $(264)^{102} + (264)^{103}$
 $= (6 + 4) = 0$.

Ex:-11 Find the total number of prime factors in the expression $(4)^n \times (7)^5 \times (11)^2$.

Solⁿ $(4)^n \times (7)^5 \times (11)^2 = (2 \times 2)^n \times (7)^5 \times (11)^2$
 $= 2^n \times 2^n \times 7^5 \times 11^2 = 2^{22} \times 7^5 \times 11^2$

Total number of prime factors $= 22 + 5 + 2 = 29$.

Ex:-12 Simplify (i) $896 \times 896 - 204 \times 204$

(ii) $387 \times 387 + 114 \times 114 + 2 \times 387 \times 114$

(iii) $81 \times 81 + 68 \times 68 - 2 \times 81 \times 68$.

HCF and LCM of Numbers

(1)

Factors and Multiples:- If a number 'a' divides another number 'b' exactly, we say that 'a' is a factor of 'b'. In this case, 'b' is called a multiple of a.

Highest Common Factor (H.C.F.) or Greatest Common Measure (G.C.M.) or Greatest Common Divisor (G.C.D.) :- The H.C.F. of two or more than two numbers is the greatest number that divides each of them exactly.

→ There are two methods of finding the H.C.F. of a given set of numbers:

(i) Factorization Method:- Express each one of the given numbers as the product of prime factors. The product of least powers of common prime factors gives H.C.F.

(ii) Division Method:- Suppose we have to find the H.C.F. of two given numbers. Divide the larger number by the smaller one. Now, divide the divisor by the remainder. Repeat the process of dividing the ~~pre~~ preceding number by the remainder last obtained till zero is obtained as remainder. The last divisor is the required H.C.F.

* Finding the H.C.F. of more than two numbers:- Suppose we have to find the H.C.F. of three numbers. Then, H.C.F. of (H.C.F. of any two) and (the third number) gives the H.C.F. of three given numbers. Similarly the H.C.F. of more than three numbers may be obtained

3) Least Common Multiple (L.C.M.) :- The least number which is exactly divisible by each one of the given numbers is called their L.C.M.

(i) Factorization Method of finding L.C.M. :- Resolve each one of the given numbers into a product of prime factors. Then, L.C.M. is the product of highest powers of all the factors.

(ii) Common Division Method (Short-Cut Method) of finding L.C.M. :- Arrange the given numbers in a row in any order. Divide by a number which divides exactly at least two of the given numbers and carry forward the numbers which are not divisible. Repeat the above process till no two of the numbers are divisible by the same number except 1. The product of ~~divisi~~ divisors and the undivided numbers is the required L.C.M. of the given numbers.

4) Product of two numbers = Product of their H.C.F. & L.C.M.

5) H.C.F. and L.C.M. of Fractions :-

$$(i). \quad H.C.F. = \frac{H.C.F. \text{ of Numerators}}{L.C.M. \text{ of Denominators}}$$

$$(ii) \quad L.C.M. = \frac{L.C.M. \text{ of Numerators}}{H.C.F. \text{ of Denominators}}$$

6) H.C.F. and L.C.M. of Decimal Fractions :- In given numbers, make the same number of decimal places by annexing zeros in some numbers, if necessary.

Ordering these numbers without decimal point. (2)
 Find H.C.F., or L.C.M. as the case may be. Now, in the result, mark off as many decimal places as are there in each of the given numbers.

Comparison of Fractions:- Find the L.C.M. of the denominators of the given fractions. Convert each of the fractions into an equivalent fraction with L.C.M. as the denominator, by multiplying both the numerator and denominator by the same number. The resultant fraction with the greatest numerator is the greatest.

Ex:-1 Find the H.C.F. of $2^8 \times 3^2 \times 5 \times 7^4$, $2^2 \times 3^6 \times 5^2 \times 7^6$, $2^3 \times 5^3 \times 7^4$.

Solⁿ The prime numbers common to given numbers are 2, 5 and 7.

$$\therefore \text{H.C.F.} = 2^2 \times 5 \times 7^2 = 980.$$

Ex:-2 Find the H.C.F. of 513, 1134, 1215.

Solⁿ

$$\begin{array}{r}
 1134 \overline{) 1215} \quad 1 \\
 \underline{1134} \\
 81 \overline{) 1134} \quad 14 \\
 \underline{81} \\
 324 \\
 \underline{324} \\
 \hline
 0
 \end{array}$$

H.C.F. of 1134 & 1215 is 81.

So required H.C.F. = H.C.F. of 513 & 81.

$$\begin{array}{r}
 81 \overline{) 513} \quad 6 \\
 \underline{486} \\
 27 \overline{) 81} \quad 3 \\
 \underline{27} \\
 0
 \end{array}$$

$\Rightarrow$ H.C.F. of given numbers = 27.

Ex:-3 Find the L.C.M. of $2^3 \times 3^2$, $3^3 \times 2^2$, $2^4 \times 5^2 \times 3 \times 7$. last

Solⁿ L.C.M. = $2^3 \times 3^3 \times 5^2 \times 7 = 37800$.

Ex:-4 Find the H.C.F. & L.C.M. of $\frac{2}{3}$, $\frac{8}{9}$, $\frac{16}{81}$ and $\frac{10}{27}$.

Solⁿ H.C.F. of given fractions = $\frac{\text{H.C.F. of } 2, 8, 16, 10}{\text{L.C.M. } 3, 9, 81, 27} = \frac{2}{81}$

L.C.M. of given fractions = $\frac{\text{L.C.M. of } 2, 8, 16, 10}{\text{H.C.F. of } 3, 9, 81, 27} = \frac{80}{3}$

Ex:-5 Find the H.C.F. and L.C.M. of 0.63, 1.05 and 2.1.

Solⁿ Making the same number of decimal places, the given numbers are 0.63, 1.05 and 2.10

Without decimal places these numbers are 63, 105, 210.

Now, H.C.F. of 63, 105, and 210 is 21.

$\therefore$ H.C.F. of 0.63, 1.05 and 2.1 is 0.21.

L.C.M. of 63, 105 and 210 is 630.

$\therefore$ L.C.M. of 0.63, 1.05 and 2.1 is 6.30.

Power Cycle

(1)

The last digit of a number of the form a^b falls in a particular sequence or order depending on the unit digit of the number 'a' and the power the number is raised to 'b'. Thus, the power cycle of a number depends on its unit digit.

Consider the power cycle of 2.

$$2^1 = 2 \quad 2^5 = 32$$

$$2^2 = 4 \quad 2^6 = 64$$

$$2^3 = 8 \quad 2^7 = 128$$

$$2^4 = 16 \quad 2^8 = 256$$

It can be observed that the unit digit gets repeated after every 4th power of 2. Hence, we can say that 2 has a power cycle of 2, 4, 6, 8 with frequency 4.

This means that, a number of the form:

• 2^{4k+1} will have the last digit as 2

• 2^{4k+2} " " " as 4

• 2^{4k+3} " " " as 8

• 2^{4k+4} " " " as 6 (where $k=0,1,2,3$)

This is applicable not only for 2 but for all numbers ending in 2. [Ex:- 12^{32} , 3452^{123}]

To find the last digit of a number raised to any power, we just need to know the power cycle of digits from 0 to 9, which is given below.

Unit digit	Power Cycle	Frequency
0	0	1
1	1	1
2	2, 4, 8, 6	4
3	3, 9, 7, 1	4
4	4, 6	2
5	5	1
6	6	1
7	7, 9, 3, 1	4
8	8, 4, 2, 6	4
9	9, 1	2

Note:- You don't need to remember the frequencies, as in every case, the frequency 4 is valid.

Ex:-1 Find the remainder when 3^{75} is divided by 5.

Sol:- $75 = 4k + 3$

$$3^{4k+3} = 3^3 = 7$$

Any number divided by 5, the remainder will be that of the unit digit divided by 5. Hence, the remainder is 2.

2/2 Concept of Unit Digit can be learned by figuring out the unit digits of all the single digit numbers from 0-9 when raised to certain powers

These numbers can be broadly classified into three categories for this purpose.

1) Digits 0, 1, 5 and 6 :- When we observe the behaviour of these digits, they all have the same unit's digit as the number itself when raised to any power i.e.

$$0^n = 0, 1^n = 1, 5^n = 5, 6^n = 6$$

Ex:- Find the unit digit of following numbers.

(i) 185^{563} Ans 5

(ii) 271^{6987} Ans 1

(iii) 156^{25369} Ans 6

(iv) $190^{654789231}$ Ans 0

2) Digits 4 and 9 :- Both these numbers have a cyclicity of only two different digits as their unit's digit. Let us take a look at how the powers of